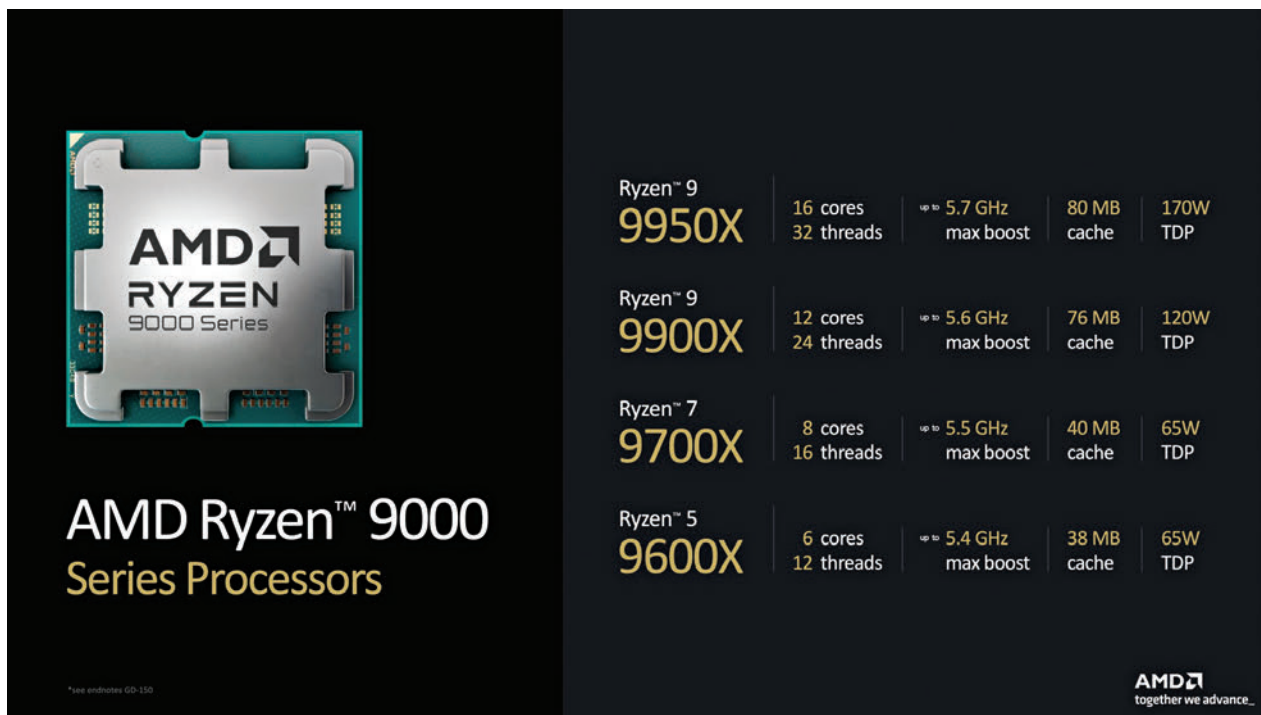




The advertisement features a large image of an AMD Ryzen 9000 Series processor on the left. To the right, a table lists the specifications for four models: Ryzen 9 9950X, Ryzen 9 9900X, Ryzen 7 9700X, and Ryzen 5 9600X. The AMD logo and slogan 'together we advance...' are at the bottom right.

Processor	Cores / Threads	Max Boost	Cache	TDP
Ryzen™ 9 9950X	16 cores / 32 threads	up to 5.7 GHz	80 MB	170W
Ryzen™ 9 9900X	12 cores / 24 threads	up to 5.6 GHz	76 MB	120W
Ryzen™ 7 9700X	8 cores / 16 threads	up to 5.5 GHz	40 MB	65W
Ryzen™ 5 9600X	6 cores / 12 threads	up to 5.4 GHz	38 MB	65W

AMD Ryzen™ 9000 Series Processors

*See AMD Ryzen 9000 Series Processor Specifications for more details.

AMD
together we advance...

simultaneous multithreading (SMT or hyper-threading). For instance, AMD's Ryzen 9000 processors provide up to 16 cores and 32 threads. Intel's 15th generation "Arrow Lake" chips combine 8 performance cores and 16 efficiency cores for a total of 24 cores/24 threads. Cores share caches – fast pools of memory that store frequently used data – and are connected to the rest of the system through memory controllers and PCI Express (PCIe) lanes.

Core Types and Architectures

Historically CPUs used identical cores. AMD continues this symmetrical design in the Ryzen 9000 series, where every core is identical and can boost up to 5.7 GHz on flagship chips. AMD's design leverages a 4 nm TSMC process, which allows high clock speeds and efficient power use. Intel's Arrow Lake / Core Ultra processors embrace a hybrid architecture with two core types:

- **Performance cores (P-cores)** focus on high single-thread speed for demanding tasks. In the Core Ultra 9 285K, P-cores run at 3.7 GHz base and 5.5 GHz boost.
- **Efficiency cores (E-cores)** handle background and multi-threaded workloads more efficiently. The same chip's E-cores operate at 3.2 GHz base and 4.6 GHz boost.

This big-LITTLE approach allows Intel to maximize performance while improving power efficiency. To further accelerate emerging workloads, Intel integrates a neural processing unit (NPU) into its Arrow Lake chips. The NPU provides 13 TOPS (tera-operations per second) of AI acceleration and works alongside the CPU and integrated GPU to run AI tasks locally. AMD counters with second-generation 3D V-Cache and built-in AI instructions; the cache is stacked beneath the processor cores, increasing on-chip memory without adding significant heat.

X3D versions of Ryzen processors deliver up to 144 MB combined cache, offering huge bandwidth for game engines and large data sets.

Clock Speeds, Cache and Power

CPUs operate at a base clock speed and can boost to higher frequencies when thermal and power budgets permit. AMD's flagship Ryzen 9 9950X has a 4.3 GHz base and 5.7 GHz boost clock, while Intel's Ultra 9 285K has P-core frequencies of 3.7 GHz base/5.5 GHz boost. These speeds translate directly into how fast a single thread can run. For multi-thread workloads, more cores and threads matter more. Caches store recent data so the processor doesn't wait for RAM. AMD's 9950X includes 80 MB of combined L2 and L3 cache, while Intel's 285K has 76 MB (40 MB L2 + 36 MB L3). In extended tasks, cache differences can reduce latency and improve throughput.

Power consumption is measured through thermal design power (TDP). AMD's standard Ryzen 9 9950X has a 170W TDP with a 230W maximum turbo; Intel's Ultra 9 285K has a 125W base power and 250W maximum turbo power. Higher power requirements demand robust cooling and a quality power supply. Mid-range chips like AMD's Ryzen 7 9700F (8 cores/16 threads, up to 5.5 GHz boost) and Ryzen 5 9500F (6 cores/12 threads) have 65W TDP and suit budget or small-form-factor builds. When choosing a CPU, ensure your power supply unit (PSU) can deliver enough wattage and your case has adequate airflow.

Choosing between Intel and AMD in 2025

Both AMD and Intel offer excellent CPUs, but they excel in different areas. In mid-2024 early benchmarks showed that AMD's